2023). Despite revealing an interplay between weight decay and large initial LR, the understanding of the corresponding dynamics remains limited. In this section, our goal is to comprehensively understand these dynamics, particularly to elucidate the difference in generalization between training with and without weight decay and using different learning rates, as observed in Fig. 2a. Given the regularization of the ℓ_2 norm of parameters, it is natural to wonder whether weight decay's improvement primarily stems from its ability to control the norm of the trained model. The experiment in Fig. 2c clearly illustrates that distinct training trajectories, while resulting in the same final ℓ_2 norm for parameters, can yield different levels of generalization stating that *the ℓ_2-norm of the learned model's parameters is inconsequential*. This observation suggests that once the norm is constrained by weight decay, the critical factor influencing the model's generalization is the subsequent choice of LR. We should note that neural networks can be explicitly made scale-invariant by means of normalization layers and small architectural changes. Li & Arora (2019) have used this setting to reveal that the training dynamics has an effective learning rate which, depending on the L2-norm of the parameters, reduces the effect of WD to merely a scheduler for the learning rate. Our analysis presents a broader perspective that does not depend on scale invariance. At the core of our examination are the unique properties of exponentially tailed loss functions, such as CE: when the data is separable and WD is not applied, the infimum of the loss is at infinity, leading to the unbounded growth of the weight norm (Ji & Telgarsky, 2019; Soudry et al., 2018). The application of WD, by inhibiting this growth, prevents the decrease of CE loss, which in turn, significantly alters the optimization dynamics.

Indeed, examining the parameter norm evolution in Fig. 2c, we notice how it rapidly decreases to stabilize within a small, approximately constant interval. Similarly, the Train CE in Fig. 2b displays a stabilization effect beyond which it cannot decrease without a reduction in the learning rate. We hypothesize that WD enables an optimization dynamic akin to that on the surface of a sphere of certain radius thus triggering the following essential mechanism:

Constraining the parameter norm hinders the decrease of the CE, thereby enabling non-vanishing noise in SGD. This allows SGD implicit regularization to unfold and steer the optimization trajectory.

Next, we empirically characterize this implicit regularization mechanism.

2.2 The noise driven process

The long-held belief that the implicit regularization property of SGD is pivotal to the generalization capabilities of Neural Networks has been a cornerstone in the field of deep learning (Keskar et al., 2016). Many theoretical studies (Blanc et al., 2020; Li et al., 2021b; Damian et al., 2021), attempting to understand this phenomenon, draw upon the essential finding that, in the case of regression and when Gaussian noise is added to the labels, the shape of the covariance of the stochastic gradients matches the shape of the Hessian. This allows Damian et al. (2021) and Pillaud-Vivien et al. (2022) to show that the trajectory of the SGD iterates closely tracks the solution of a regularized problem. In our analysis, we conjecture a similar result; the dynamics of SGD with CE, closely track a regularized process. The important difference in our statement is that unlike Blanc et al. (2020) and Damian et al. (2021), we do not need to add noise to the labels at each iteration. Instead, weight decay, in combination with large-LR induces a label noise-like behavior via loss stabilization (Andriushchenko et al., 2023). To better understand the interplay between weight decay, loss stabilization and the noise of SGD, it is convenient to consider the binary classification case where $y_i \in \{0, 1\}$. We also define the Jacobian of the network as $J(x_i, \mathbf{w}) := \nabla h_{\mathbf{w}}(x_i) \in \mathbb{R}^p$ and its norm averaged across the dataset: $\left\|J(\mathbf{w})\right\|_F^2 = \frac{1}{N}\sum_{i=1}^N \operatorname{Tr}\left(\nabla h_{\mathbf{w}}(x_i)\nabla h_{\mathbf{w}}(x_i)^\top\right)$. Denoting the noise of the gradient by $g_t = \nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}_t) - \nabla_{\mathbf{w}}\ell\left(y_{i_t}, h(\mathbf{w}_t, x_{i_t})\right)$, the SGD update in equation 1 becomes:

$$\mathbf{w}_{t+1} = (1 - \eta\lambda)\mathbf{w}_t - \eta\nabla\mathcal{L}(\mathbf{w}_t) + \eta g_t\,. \tag{2}$$

Furthermore, we consider a Gaussian approximation of the SGD noise, matching the first and second moment of g_t. A substantial body of research has built upon this approximation and verified its validity (Li et al., 2020, 2021a; Smith et al., 2020; Xie et al., 2020; Li et al., 2021b). In particular Li et al. (2021a) demonstrated how modelling the SGD noise by a Gaussian is sufficient to understand its generalization. The mean is zero due to the unbiased mini-batch gradients: $\bar{g}(\mathbf{w}_t) := \mathbb{E}[g_t] = 0$

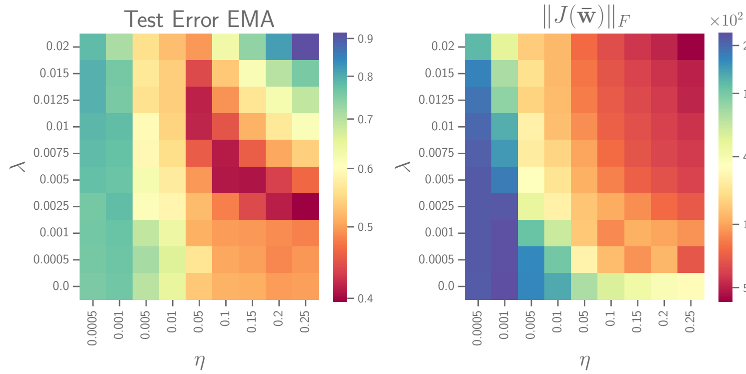


Figure 3: **Resnet18 on Tiny-ImageNet.** Heatmap of the test error and Jacobian norm for the EMA for all the different combinations of η and λ.

whereas for the second moment:

$$\Sigma_{\mathbf{w}_t} \coloneqq \frac{1}{N}\sum_{i=1}^{N} \nabla \ell_i(\mathbf{w}_t)\nabla \ell_i^\top(\mathbf{w}_t) - \nabla\mathcal{L}(\mathbf{w}_t)\nabla\mathcal{L}^\top(\mathbf{w}_t) \approx \frac{1}{N}\sum_{i=1}^{N} \nabla \ell_i(\mathbf{w}_t)\nabla \ell_i^\top(\mathbf{w}_t)$$
$$\approx \frac{1}{N}\sum_{i=1}^{N} (\ell_i'(\mathbf{w}_t))^2 \nabla h_{\mathbf{w}_t}(x_i)\nabla h_{\mathbf{w}_t}(x_i)^\top \approx \frac{1}{N}\sigma^2_{\eta,\lambda}(\mathbf{w}_t)\sum_{i=1}^{N}\nabla h_{\mathbf{w}_t}(x_i)\nabla h_{\mathbf{w}_t}(x_i)^\top . \quad (3)$$

In equation 3, we consider $\nabla\mathcal{L}(\mathbf{w}_t)\nabla\mathcal{L}^\top(\mathbf{w}_t)$ negligible compared to $\nabla \ell_i(\mathbf{w}_t)\nabla \ell_i^\top(\mathbf{w}_t)$ as the gradient noise variance dominates the square of the gradient noise mean. This fact has been used in previous works (Mori et al., 2022; Zhu et al., 2018; Jastrzebski et al., 2017) and in particular Jastrzebski et al. (2017) and Saxe et al. (2019) empirically verify it. Finally, we assume that the first derivative is approximately constant across all datapoints: $\ell_i'(\mathbf{w}_t) \approx \ell_j'(\mathbf{w}_t)\ \forall i,j$ and denote this common quantity as $\sigma_{\eta,\lambda}(\mathbf{w}_t)$. This last approximation is referred to as "decoupling approximation" and has been empirically verified for classification (Mori et al., 2022). Furthermore, in App. C.6 we performed additional experiments to verify the decoupling approximation by comparing the spectrum of the SGD covariance with and without this approximation during the large LR phase.

Altogether, these considerations lead to the following formulation of the SGD update:

$$\mathbf{w}_{t+1} \approx (1-\eta\lambda)\mathbf{w}_t - \eta\nabla\mathcal{L}(\mathbf{w}_t) - \frac{\eta}{N}\sigma_{\eta,\lambda}(\mathbf{w}_t)\sum_{i=1}^{N}\nabla h_{\mathbf{w}_t}(x_i)\,\xi_i^t, \qquad \text{where } \xi_i^t \sim \mathcal{N}(0,\mathbb{I}). \quad (4)$$

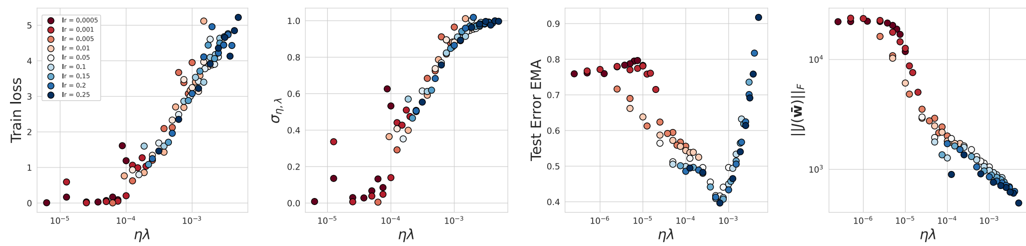
This series of approximations, allows us to define the quantity $\sigma_{\eta,\lambda}(\mathbf{w}_t)$, which has a fundamental role in the characterization of the training dynamics. We refer to it as *the scale of the noise* because it regulates its intensity. Although η and λ influence the noise scale indirectly through the trajectory of$\mathbf{w}_t$, we explicitly highlight this dependence to emphasize our objective: to characterize the influence of WD and LR on the stochastic dynamics of SGD. We develop this characterization building upon the connection between the Jacobian of the network and the covariance of the SGD noise, which motivates the introduction of the following implicit regularization mechanism:

Conjecture 1. *Consider the algorithm in Eq. 1 with $\mathbf{w}_0$ initialized from a distribution $\mu_0\left(\mathbb{R}^{(p)}\right)$. For any input x, let $\mathbf{w}_t, h(\mathbf{w}_t, x)$ be the random variables that denote the iterate at time t and its functional value. The stochastic process $(h(\mathbf{w}_t, x))_{t\in\mathbb{N}}$ converges to the stationary distribution $\mu^\infty_{\eta,\lambda}(x)$ with mean $\bar{\mu}_{\eta,\lambda}(x) = h\left(\mathbf{w}^*_{\eta,\lambda}, x\right)$ for which $\mathbf{w}^*_{\eta,\lambda}$ is a first-order stationary point of the following regularized loss:*

$$\bar{\mathcal{L}}_\lambda(\mathbf{w}) \coloneqq \mathcal{L}_\lambda(\mathbf{w}) + \eta\sigma^2_{\eta,\lambda}\left\|J(\mathbf{w})\right\|_F^2. \quad (5)$$

The conjecture illustrates how varying noise levels correspond to distinct processes, wherein the mean of each solves a unique regularized problem. Moreover, it describes how each noise level determines the strength of the regularization. Using the mean to formulate the conjecture is a natural choice; even at stationarity,[3] the values of the loss $\mathcal{L}(\mathbf{w}_t)$ and the regularizer $\left\|J(\mathbf{w})\right\|_F^2$ would be dominated

[3]Assuming the existence of a stationary distribution, the iterates $\mathbf{w}_t$ are eventually realizations from it.



(a) (b) (c) (d)

Figure 4: **Resnet18 on Tiny-ImageNet.** Training for 200 epochs with different η and λ; the scale of the noise monotonically increases with the train loss and $\eta \times \lambda$ Fig. 4a, 4b. The test error instead, presents an optimal value of $\eta \times \lambda$ Fig. 4c while the Jacobian norm decreases monotonically Fig. 4d.

by the noise. To unveil the existence of an implicit regularization and to analyze the evolution of the distribution, we need to look at its summary statistics, in this case, the mean. While insights from Langevin dynamics suggest employing learning rate annealing to converge towards the mean of the stationary distribution, this approach introduces additional complexities which we discuss in Section 2.3. We instead consider an exponential moving average (EMA) $(\bar{\mathbf{w}}_t)_{t\geq 0}$ of the SGD iterates with parameter $\beta = 0.999$.

The most important implication of the conjecture is that the strength of the regularization $\sigma_{\eta,\lambda}$ depends on both the LR η and the WD parameter λ. Our experiments in Fig. 3, provide empirical validation for this conjecture. When trained with different combinations of η and λ, the EMA converges to models with different test performances. When fixing λ there exists an optimal value of learning rate η which gives the best test performance while the Jacobian norm monotonically increases. A similar picture can be drawn when fixing the learning rate.

Therefore, given two solutions $\bar{\mu}_{\eta_l,\lambda_l}$ and $\bar{\mu}_{\eta_s,\lambda_s}$ for which Test error$(\bar{\mu}_{\eta_l,\lambda_l}) <$ Test error$(\bar{\mu}_{\eta_s,\lambda_s})$; the difference in their performances can be explained with the difference in their regularization strengths $\sigma_{\eta,\lambda}$. The solution $\bar{\mu}_{\eta_l,\lambda_l}$ benefits from better regularization and therefore endows better generalization properties. Furthermore, the heatmap for test error in Fig. 3 indicates that the minimum error is not achieved by a single combination of η and λ, but rather along a contour where their product $\eta \times \lambda$ appears to be constant. This observation suggests an optimal trade-off between the learning rate and weight decay parameter, characterized by a curve in the parameter space where their product remains constant. Characterizing this relationship might reveal a useful tool which practitioners can adopt to optimally select values of weight decay and learning rate. Fig. 4c, 4d confirm that the product $\eta\lambda$ is the quantity controlling the regularization; combinations of η and λ with the same product show similar test performances and Jacobian norm. For Tiny-Imagenet, the test error exhibits an optimal value for $\eta\lambda \sim 0.005$ where increases beyond this point lead to *over-regularization* and decreases result in *under-regularization*. Simultaneously, the Jacobian norm $\left\|J\right\|_F$ consistently exhibits a monotonically decreasing trend.

To better understand the properties of the noise scale during training, we can observe in Fig. 4a, 4b how higher values of the training loss given by larger $\eta\lambda$ correspond to higher values of the noise scale $\sigma_{\eta,\lambda}$. The latter is measured by computing the Frobenius norm of the first derivative of the loss with respect to the predictions[4] averaged over all training datapoints $\frac{1}{N}\sum_{i=1}^{N}\left\|\ell_i'(\mathbf{w})\right\|_F$. The scale of the noise and therefore the strength of the regularization vanish when $\mathcal{L}(\mathbf{w}) \approx 0$ which happens for small values of $\eta\lambda$. Therefore, WD combined with a large LR stabilizes the CE to a larger value, preventing the noise from vanishing and regulating the implicit regularization. This fact further confirms Conjecture 1, demonstrating that with smaller noise scales, the mean (EMA) tends towards a point where the Jacobian's norm is higher, compared to trajectories with larger noise scales. To further validate our conjecture, we created snapshot ensembles in the ResNet18 on CIFAR-10 setting, by averaging in function space along the SGD trajectory every 10 epochs for the combinations of learning rate (LR) and weight decay (WD) considered. To assess whether the mean of the stationary distribution in function space aligns closely with the EMA, where the Jacobian norm is regularized, we compared the performance of snapshot ensembles with that of the EMA. Additionally, we computed the Total Variation Distance between the softmax outputs of the ensemble and the EMA. The results are reported in App. C.7 and show a strong alignment.

[4] In the multi-class setting $\ell_i'(\mathbf{w}_t) \in \mathbb{R}^c$; to quantify the scale, we compute its Frobenius norm.